Incomplete Markets II

Endogenous Incomplete Markets: Limited Commitment

QEM — Macroeconomics III

Contents

1 Motivation: Why Endogenous Incompleteness?

In **Incomplete Markets I** (Bewley–Huggett–Aiyagari), market incompleteness was *imposed*: only a single risk-free bond was assumed to exist, with no explanation of why richer insurance markets are absent.

There are three approaches to modelling incomplete markets:

Approach	**Why markets are incomplete**	**Example**
Exogenous	Assumed — only a bond exists	Bewley, Aiyagari
Limited Commitment	Agents cannot commit to future contracts	This section
Private Information	Moral hazard / adverse selection	—

Intuition

In the real world, full insurance markets do not exist. The question is *why*. Endogenous incomplete markets derive this from a primitive friction rather than an assumption.

2 Setting

- **Infinitely lived, risk-averse households.** Preferences: $\sum_{t=0}^{\infty} \beta^t u(c_t)$, with u strictly concave.
- **Endowment risk.** Each household receives a stochastic endowment $y_t \in \{y_l, y_m, y_h\}$ each period, drawn from an exogenous Markov process. There is no production — only endowments.
- **Common knowledge.** Probabilities (ex ante) and realizations (ex post) are observable by everyone. There is *no* private information — if your income is low, everyone knows it.
- **No commitment.** Households cannot commit to honouring future contracts. If it is advantageous to default tomorrow, they will.
- **No saving or borrowing.** There is no bond market. The *only* way to smooth consumption is through insurance contracts between households.

Notation.

- s_t: realization of the shock at period t (a single state, like a photograph).
- $s^t = (s_0, s_1, \ldots, s_t)$: the full history of shocks up to t (the entire film until now).
- $\pi(s^t)$: unconditional probability of history s^t.
- $\pi(s^r \mid s^t)$: probability of future history s^r conditional on being at s^t today ($r \geq t$).

Intuition

Why does s^t vs. s_t matter? The contract can use the full history s^t to assign consumption ("you had bad luck last period, so you get more today"). Autarky only depends on the current state s_t, because once you leave the contract, only your current income matters — the Markov property means the past is irrelevant.

3 Objective: Transfers that Respect Participation Constraints

We want to find a **transfer arrangement** that:

1. **Reduces consumption risk** — smooths consumption across good and bad states.
2. **Respects participation constraints (PCs)** — no household ever prefers to walk away from the contract.

Key Insight

The fundamental tension: larger transfers give better insurance, but if transfers are too large, the household making the payment prefers to default. The equilibrium contract is the *best* insurance compatible with voluntary participation.

4 What is Consumption Smoothing?

Consumption smoothing means **reducing the variability of consumption** relative to what it would be under autarky.

Example. Suppose income alternates: $y_h = 100$, $y_l = 20$.

Regime	**Consumption (good period)**	**Consumption (bad period)**
Autarky	100	20
Perfect insurance	60	60
Limited commitment	70	50

With limited commitment, the contract provides *partial* insurance: consumption still varies, but less than income does.

Intuition

Because u is concave, Jensen's inequality gives $u(\bar{c}) > \mathbb{E}[u(c_t)]$. Households prefer a stable consumption stream to a volatile one with the same mean. Smoothing consumption raises welfare even when average resources are unchanged.

5 The Social Planner's Problem

5.1 Without Participation Constraints (First Best)

The planner maximizes weighted welfare subject only to resource feasibility:

$$\max_{\{c_i(s^t)\}} \sum_i \lambda_i \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t)\, u_i\big(c_i(s^t)\big) \tag{1}$$

subject to

$$\sum_i c_i(s^t) \;\leq\; \sum_i y_i(s_t), \qquad \forall\, s^t. \tag{2}$$

where:

- $\lambda_i \geq 0$: **Pareto weight** of household i — how much the planner values household i's utility. Determined at $t = 0$ by initial wealth.
- The resource constraint depends on s_t (current income), not the full history s^t, because income is Markovian.
- Consumption $c_i(s^t)$ can depend on the full history s^t, because the planner can use past information to allocate resources.

5.2 Participation Constraints (Second Best)

Since households cannot commit, the planner must also ensure that in every history s^t and for every household i, staying in the contract is weakly preferred to autarky:

$$\underbrace{\sum_{r=t}^{\infty} \sum_{s^r} \beta^{r-t}\, \pi(s^r \mid s^t)\, u_i(c_i(s^r))}_{\text{value of staying in the contract from } s^t} \;\geq\; \underbrace{U_i^{\text{aut}}(s_t)}_{\text{value of autarky}}\;, \qquad \forall\, s^t,\, \forall\, i. \tag{3}$$

Value of autarky. If household i exits at state s_t, it consumes its own endowment forever. By the Markov property, this value depends only on the current state s_t:

$$U_i^{\text{aut}}(s_t) = \sum_{r=t}^{\infty} \sum_{s^r} \beta^{r-t}\, \pi(s^r \mid s_t)\, u_i(y_i(s_r))\,. \tag{4}$$

Key asymmetry.

- The *left-hand side* depends on the full history s^t: the contract can use past events to set consumption.
- The *right-hand side* depends only on s_t: once you leave, only your current state matters.

When is the PC typically binding? When household i has a *high* income draw ($s_t = y_h$) and the contract requires a large transfer. The household is tempted to keep its income and default. The PC is never binding for the household in the bad state, because that household is *receiving* a transfer.

6 The Lagrangian and Recursive Formulation

6.1 Full Lagrangian

Attaching multiplier $\beta^t \pi(s^t)\, \mu_i(s^t) \geq 0$ to the PC of household i at history s^t, and $\delta(s^t)$ to the resource constraint at s^t:

$$\begin{aligned} L = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) \Bigg[& \sum_i \lambda_i\, u\big(c_i(s^t)\big) \\ & + \sum_i \mu_i(s^t) \left(\sum_{v=t}^{\infty} \sum_{s^v} \beta^{v-t}\, \pi(s^v \mid s^t)\, u(c_i(s^v)) - U_i^{\text{aut}}(s_t) \right) \\ & + \delta(s^t) \sum_i \big(y_i(s^t) - c_i(s^t)\big) \Bigg]. \end{aligned} \tag{5}$$

The Lagrange multiplier $\mu_i(s^t)$:

- $\mu_i(s^t) = 0$: the PC of household i is *slack* at s^t — the household strictly prefers the contract; no adjustment needed.
- $\mu_i(s^t) > 0$: the PC is *binding* at s^t — the household is exactly indifferent between staying and leaving; the planner must give it more consumption to retain it.

6.2 Recursive Form

Using Bayes' rule ($\pi(s^r) = \pi(s^r \mid s^t)\, \pi(s^t)$) and Abel's summation formula to reorganize the double sum, the Lagrangian reduces to:

$$\boxed{L = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) \left[\sum_i M_i(s^t)\, u\big(c_i(s^t)\big) - \mu_i(s^t)\, U_i^{\text{aut}}(s_t) + \delta(s^t) \sum_i \big(y_i(s^t) - c_i(s^t)\big) \right]} \tag{6}$$

where the **cumulative Pareto weight** $M_i(s^t)$ is defined by:

$$M_i(s^t) \;=\; M_i(s^{t-1}) + \mu_i(s^t), \qquad M_i(s^{-1}) = \lambda_i. \tag{7}$$

Intuition

$M_i(s^t)$ accumulates the initial Pareto weight λ_i plus all binding PC adjustments up to history s^t. Every time household i's PC binds, its weight permanently increases. Weights *never decrease.*

7 First-Order Condition and the Ratio of Marginal Utilities

7.1 FOC

Taking the derivative of L with respect to $c_i(s^t)$:

$$M_i(s^t)\, u'\big(c_i(s^t)\big) = \delta(s^t), \qquad \forall\, i. \tag{8}$$

For two households i and k, dividing their FOCs:

$$\frac{u'\big(c_i(s^t)\big)}{u'(c_k(s^t))} = \frac{M_k(s^t)}{M_i(s^t)} \equiv X(s^t). \tag{9}$$

Expanding M:

$$X(s^t) = \frac{\lambda_k + \mu_k(s^0) + \cdots + \mu_k(s^t)}{\lambda_i + \mu_i(s^0) + \cdots + \mu_i(s^t)}. \tag{10}$$

7.2 Two Cases

Case	**Condition**	**Implication**
Perfect risk sharing	$\mu_i(s^t) = 0\ \forall\, i,\ s^t$	$X(s^t) = \lambda_k/\lambda_i$ — constant across time and states
Limited commitment	$\mu_i(s^t) > 0$ when PC binds	$X(s^t)$ changes — partial and history-dependent insurance

Key Insight

$X(s^t)$ **is the key co-state variable** (Marcet–Marimon, 1998). It summarizes the entire relevant history in a single number, enabling a fully recursive representation of the problem.

8 Recursive Value Function and the Updating Rule

8.1 Value Function

Because X_{t-1} summarizes all relevant history, the value function can be written recursively as:

$$V_i(s_t, x_{t-1}) = u(c_i(s_t, x_{t-1})) + \beta \sum_{s_{t+1}} \pi(s_{t+1} \mid s_t)\, V_i(s_{t+1},\, x_t(s_{t+1}, x_{t-1}))\,. \tag{11}$$

The individual state is now (s_t, x_{t-1}): current income and the ratio inherited from the past.

8.2 The LTW Updating Rule

Each state s admits a **participation-compatible interval** $[\underline{x}^s, \bar{x}^s]$ for x. The updating rule is:

$$x_t(s_t, x_{t-1}) = \begin{cases} \bar{x}^{s_t} & \text{if } x_{t-1} > \bar{x}^{s_t} \\ x_{t-1} & \text{if } x_{t-1} \in [\underline{x}^{s_t},\, \bar{x}^{s_t}] \\ \underline{x}^{s_t} & \text{if } x_{t-1} < \underline{x}^{s_t} \end{cases} \tag{12}$$

Interpretation:

- If x_{t-1} is inside the interval: no PC is binding — x stays unchanged, contract continues as before.
- If x_{t-1} is outside the interval: some PC binds — x is pushed to the nearest boundary, adjusting consumption to retain the household.

9 Numerical Example

Setup. Two households, $u = \ln$, $x_{-1} = 1$. Income takes three values: low = 2, medium = 5, high = 8.

Time 0: (5, 5) — **both medium**

$1 \in [\underline{x}^{mm}, \bar{x}^{mm}] \;\Rightarrow\; x_0 = x_{-1} = 1$.

No PC is binding. No transfers are made. Both households consume their endowment of 5.

Time 1: (8, 2) — **household 1 rich, household 2 poor**

$1 \notin [\underline{x}^{hl}, \bar{x}^{hl}]$, so $x_1 = \underline{x}^{hl} > 1$.

Household 1's PC is binding: it is tempted to default and keep its entire income. The planner pushes x to $\underline{x}^{hl}$, meaning household 1 makes a transfer (say, 2 units). This is *less* than the transfer under perfect risk sharing (3 units), because the PC constrains how much can be extracted.

Time 2: (5, 5) — **both medium again**

Now $x_1 = \underline{x}^{hl} > 1$. Two sub-cases arise:

- **Case A:** $\underline{x}^{hl} \in [\underline{x}^{mm}, \bar{x}^{mm}]$. Then $x_2 = x_1 > 1$. No new PC is binding. Because $x > 1$, household 2 now has higher relative weight and *receives* a transfer (say, 1 unit) even though both incomes are equal.
  ⇒ **History dependence:** the contract "remembers" that household 1 was rich last period.
  ⇒ **Quasi-credit element:** household 2 is effectively collecting on an implicit debt.
- **Case B:** $\underline{x}^{hl} \notin [\underline{x}^{mm}, \bar{x}^{mm}]$. Then $x_2 < x_1$. The history is partially forgotten.

Time 3: (2, 8) — **household 1 poor, household 2 rich**

$x_2 \notin [\underline{x}^{lh}, \bar{x}^{lh}] \;\Rightarrow\; x_3 = \bar{x}^{lh}$.

Now household 2 is the rich one and its PC binds. The co-state variable jumps to $\bar{x}^{lh}$, wiping out the implicit credit accumulated in favour of household 2.

⇒ **Amnesia:** when the roles reverse, all past "debt" is permanently forgotten.

Result

Summary of the dynamics of x_t:

- **PC not binding:** x stays constant — history is preserved.
- **Rich household's PC binds:** x rises — the rich household gets a permanent upgrade in its Pareto weight.
- **Poor-turned-rich household's PC binds:** x resets — the economy has amnesia.

Insurance under limited commitment is **partial, persistent, and asymmetric**: implicit debts accumulate but can be erased when fortune reverses.

10 Main Conclusions

1. **Limited commitment generates endogenously incomplete insurance.** Even though information is symmetric and contracts are voluntary, full risk sharing is unachievable because large transfers violate participation constraints.

2. **The co-state variable $X(s^t)$ summarizes history.** The ratio of Pareto weights is sufficient to track all relevant past events, enabling a recursive representation of an inherently history-dependent problem.

3. **The contract has memory and a credit-like structure.** Past good luck creates obligations; the contract enforces partial repayment until roles reverse.

4. **Amnesia when roles reverse.** When the formerly rich household becomes poor, accumulated implicit debts are reset — a direct consequence of the no-commitment friction.